

knee and ankle joints that operate in unison. Sensors monitor the user's motion. Microprocessors use this data to predict what the person is trying to do and operate the device in ways that facilitate these movements.

The Vanderbilt leg weighs about 4 kg and can operate for three days of normal activity, or thirteen to fourteen kilometres of continuous walking on a single charge. If it senses that the user is starting to stumble, it lifts up to clear any obstruction and plant the foot on the floor.

## Researchers combine quantum processor with quantum memory

Matteo Mariantoni and his team of scientists at the University of California, Santa Barbara, have combined a quantum processor with memory mechanisms on a single chip. The quantum integrated circuit implements the quantum von Neumann architecture. In this architecture, a long-lived quantum random-access memory can be programmed using a quantum central processing unit, all constructed on a single chip, providing the key components for a quantum version of a classical computer.

The hardware is based on superconducting quantum circuits, and must be cooled to very low temperatures to display quantum behaviour. The architecture represents a new paradigm in quantum information processing, and shows that quantum large-scale integration is within reach.

The quantum integrated circuit includes two quantum bits (qubits), a quantum communication bus, two bits of quantum memory and a resetting register comprising a simple quantum computer. The architecture explores the possibility of writing quantum information to memory, while simultaneously performing other quantum calculations.

## Electric cars with solar-powered roof

Daimler's concept car Smart Forvision combines a futuristic, lightweight

design with technologies relating to energy efficiency and temperature management. The all-electric concept car has a transparent solar-panel roof that helps generate power to keep the cooling fans running even when the car is stationary. It also has a see-through dashboard along with clear OLEDs for internal lighting at night.

The company worked with BASF on this project, which helped in developing all-plastic wheels, along with a super-thin heat shield.

## World's smallest electric motor

Researchers at Tuft University have developed an even smaller electric motor—molecular electric motor. According to the *Guinness Book of World Records*, a 200nm motor is the world's smallest. But now this molecular electric motor, which is 1nm across, will replace it to become the world's smallest.

The researchers took the help of a scanning tunnelling microscope and were able to land on a single molecule, measure it and spin it. This microscope provided electric charge to butyl methyl sulfide molecule that was placed on a copper plate. The interconnected atoms work as interlocking cogs, so when one molecule rotates the adjoining molecules too start rotating alongside. The size of the motor can be gauged from the fact that a single strand of human hair is 60,000nm

wide.

This motor can find applications in domains ranging from medicine to engineering. The tunnelling microscope and low temperature scanning played a major role in its development.

## Magic mirror

A New York Times' research and development lab team has developed a mirror that also has the ability to scan RFID tags and bring up information about a sample product. For instance, if you have a bottle of prescription medication tagged with an RFID chip, holding it up to the mirror will show important information such as directions for use, time to take the medication and doctor's information.

The mirror uses Philips Mirror TV, whose display is tied to a Microsoft Kinect sensor that sits at the top of the mirror and reads physical cues from the user. It allows users to check the latest headlines in *The New York Times* and updates on Facebook and Twitter, and also read e-mails.

## Energy-harvesting chips to power batteries

New York-based MicroGen is developing energy-harvesting chips that will recharge batteries and other wireless sensors. These will be useful for devices that need frequent battery replacements.

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